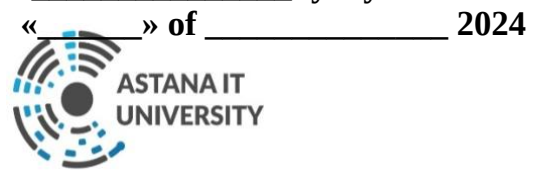


«Approved»
By the Dean
Syzdykova Z.A.



SYLLABUS

Academic Year 2024–2025

1. General information	
Course Code	GOOP 3210
Course Title	Deep and Reinforcement Learning
Degree cycle (level)/ major	Bachelor
Year, term	3
Number of Credits	5
Language of Delivery	English
Prerequisites	Machine Learning and deep learning basics
Postrequisites	Deep learning courses
Lecturer	Darkhan Zholtyaev, PhD in robotics engineering, Assistant Professor E-mail: darkhan.zholtayev@astanait.edu.kz , C1.1.332 Astana IT University, Department of Computational and Data Science
2. Goals, objectives and learning outcomes of the course	
Course description	The course explores foundational algorithms and advanced methodologies in deep reinforcement learning (DRL), emphasizing their applications and impact across various domains of science and technology. Students will gain hands-on experience with state-of-the-art DRL techniques, leveraging neural networks to model complex environments and solve decision-making problems. As an outcome of mastering the course, students will be equipped to design, train, and evaluate DRL models, visualize policy behaviors, analyze performance, and present results effectively, bridging theory with real-world applications.
Course objectives	Students will study the following key concepts: - Fundamentals of reinforcement learning and deep reinforcement learning; - Neural network architectures for reinforcement learning; - Policy-based and value-based methods; - Exploration-exploitation trade-offs; - Model-free and model-based learning approaches; - Training and evaluation workflows in DRL; - Applications of DRL in robotics, gaming, and other domains; - Integration of DRL with computer vision techniques; - Practical challenges and solutions in DRL implementations.

Course goals	The course aims to equip students with both theoretical and practical knowledge in the field of deep reinforcement learning, focusing on the development of algorithms capable of learning and making decisions autonomously in complex environments. Students will gain expertise in creating, implementing, and optimizing DRL models to solve real-world problems across various domains.
Skills & Competences	Skills & Competences: Deep Reinforcement Learning After completing this course, students should demonstrate competency in the following skills: • Proficiency in tools and frameworks for implementing and training deep reinforcement learning models (e.g., TensorFlow, PyTorch, OpenAI Gym). • Ability to design and evaluate DRL algorithms, including policy optimization and value function approximation. • Competence in analyzing and preprocessing data for DRL applications, including environment modeling and state representation. • Expertise in applying DRL to solve problems in robotics, gaming, and other real-world scenarios. • Capability to visualize and interpret DRL model performance, detect anomalies, and optimize learning strategies.
Course Learning Outcomes	Having completed the course, students will be able to: • Apply practical programming skills to implement deep reinforcement learning algorithms using frameworks such as PyTorch or TensorFlow. • Explain the general methodological principles of developing DRL models and integrating them into complex systems. • Design, train, and evaluate DRL models for solving tasks in dynamic and uncertain environments. • Develop system-level programs for interfacing DRL agents with real-world systems, such as robotics platforms or simulation environments. • Optimize DRL algorithms for performance and efficiency in practical applications.
Methods of Assessment	The expected learning outcomes for the course will be assessed through graded activities and ungraded activities. The graded activities include exams, homework, practice, SIS, and mid-terms, end-terms. The ungraded activities will be used to monitor your progress. A variety of these ungraded assessment techniques may be employed, including problems to be completed during class, direct questioning of students, answering questions in class and discussions during office hours.

Reading List	Basic Literature: 1. Book Title: <i>Reinforcement Learning: An Introduction (Second Edition)</i>, Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018 2. Book Title: <i>Deep Learning with Python (Second Edition)</i>, Author: François Chollet, Publication Year: 2021 3. Book Title: <i>Hands-On Reinforcement Learning with Python: Master Reinforcement and Deep Reinforcement Learning Using OpenAI Gym and TensorFlow</i>, Author: Sudharsan Ravichandiran, Publication Year: 2018 4. Book Title: <i>Applied Reinforcement Learning with Python: With OpenAI Gym, TensorFlow, and Keras</i>, Author: Taweh Beysolow II, Publication Year: 2019
Resources	Online journals, article, papers, books and internet resources.
Course policy	Course and university policies include: Attendance: Attendance is not allocated any grading points in the marking scheme but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 80% to get admitted to the examination rubric. In case a student misses 20% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more, he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. Class work: The duration of each lecture and practical lesson is 50 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student’s misconduct, the student can be dispensed from the classes. Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and

generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. By the policy of this course, students who come late to class for more than 5 minutes are not allowed to get in to class and consequently, they will be marked as “absent” for the specific hour.

The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround.

Homework/Assignments:

Late assignments: Most assignments will be discussed in class on the due date, therefore late assignments will not receive credit. It is expected that all work will be submitted on time. Failure to pass assignments in on time will result in 0% for the assignment. In other words, no late submissions are allowed. All gradings are based using a percentage grading scale. In the event of some extraordinary event, students should notify the teacher and request an extension of the deadline. If approved, a new date will be given to the student depending upon the circumstances.

Final exam: The final exam for course will be taken in the form of a computer test organized with the Learning Management System (moodle.astanait.edu.kz).

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

- 1 Submitting work that is not your own papers, assignments, or exams;
- 2 Copying ideas, words, or graphics from a published or unpublished source without appropriate citation.
- 3 Submitting or using falsified data;
- 4 Submitting the same work for credit in two courses without prior consent of both instructors.

Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university.

Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz).

Contacting the Instructor (Teacher): The easiest and most reliable way to get in touch with the teacher is by telegram and e-mail. Students must feel free to send email if you have a question related to the course. The teachers will respond as soon as they can but not always instantaneously. Besides that, students are also welcome to arrange a one-to-one meeting with the teacher by their office during office hours to discuss the class.

* Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Weeks	Topic and Content	Lectures (H/W)	Practice sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Introduction to Reinforcement Learning (RL) - Overview of RL and key concepts (agent, environment, reward, policy, value). - Real-world applications of RL	2	3	1	1
2	Multi-Armed Bandit Problem - Introduction to the exploration-exploitation dilemma. ϵ-Greedy, Upper Confidence Bound (UCB), and Thompson Sampling.	2	3	1	1
3	Markov Decision Processes (MDP) - Introduction to MDPs: states, actions, rewards, transitions. - Bellman equations.	2	3	1	1
4	Dynamic Programming (DP) - Policy evaluation, policy improvement, and policy iteration. - Value iteration	2	3	1	1
5	Monte Carlo Methods - Introduction to Monte Carlo methods for RL. - Exploring policy evaluation with sampling.	2	3	1	1

Mid- term					
6	Temporal Difference (TD) Learning - Difference between TD, DP, and Monte Carlo methods. - TD(0) and SARSA (State-Action-Reward-State-Action).	2	3	1	1
7	Q-Learning and Off-Policy Methods - Introduction to Q-Learning as an off-policy method. - Comparison with SARSA.	2	3	1	1
8	Deep Q-Networks (DQN) Introduction to deep reinforcement learning and DQN. Discuss neural network architectures for Q-function approximation.	2	3	1	1
9	Policy Gradient Methods Introduction to policy gradients and REINFORCE algorithm. Difference between value-based and policy-based methods.	2	3	1	1
10	Advanced Topics and Project Showcase - Overview of advanced RL topics: Actor-Critic, PPO, A3C. - Real-world RL applications and ethical considerations.	2	3	1	1
End-term					
	Total hours: 150	20	30	10	90

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Introduction to Reinforcement Learning (RL) Overview of RL and key concepts (agent,	9	<i>Reinforcement Learning: An Introduction (Second Edition), Authors: Richard S. Sutton and</i>	•Implement a basic environment using OpenAI Gym.

	environment, reward, policy, value). Real-world applications of RL.		Andrew G. Barto, Publication Year: 2018	Experiment with a random policy to familiarize with environment interactions.
2	Multi-Armed Bandit Problem - Introduction to the exploration-exploitation dilemma. ϵ-Greedy, Upper Confidence Bound (UCB), and Thompson Sampling.	9	<i>Reinforcement Learning: An Introduction (Second Edition)</i>, Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018	- Code ϵ-Greedy and UCB for a multi-armed bandit problem. - Visualize and analyze the outcomes of different strategies.
3	Markov Decision Processes (MDP) - Introduction to MDPs: states, actions, rewards, transitions. - Bellman equations.	9	<i>Reinforcement Learning: An Introduction (Second Edition)</i>, Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018	- Implement a small grid world in Python. - Calculate value functions manually and visualize state values.
4	Dynamic Programming (DP) - Policy evaluation, policy improvement, and policy iteration. - Value iteration.	9	<i>Reinforcement Learning: An Introduction (Second Edition)</i>, Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018	- Code policy and value iteration in the grid world environment. - Experiment with convergence and policy optimization.
5	Monte Carlo Methods Introduction to Monte Carlo methods for RL.	9	<i>Reinforcement Learning: An Introduction (Second Edition)</i>, Authors:	Implement Monte Carlo

	Exploring policy evaluation with sampling.		Richard S. Sutton and Andrew G. Barto, Publication Year: 2018	control in the grid world. Analyze and compare with DP methods.
6	Temporal Difference (TD) Learning - Difference between TD, DP, and Monte Carlo methods. - TD(0) and SARSA (State-Action-Reward-State-Action).	9	<i>Reinforcement Learning: An Introduction (Second Edition)</i> , Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018	- Implement TD(0) and SARSA on a simple problem. - Analyze learning stability and speed.
7	Q-Learning and Off-Policy Methods - Introduction to Q-Learning as an off-policy method. - Comparison with SARSA.	9	<i>Reinforcement Learning: An Introduction (Second Edition)</i> , Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018	- Code Q-learning for the grid world. - Experiment with learning rates and exploration rates.
8	Deep Q-Networks (DQN) - Introduction to deep reinforcement learning and DQN. - Discuss neural network architectures for Q-function approximation.	9	<i>Reinforcement Learning: An Introduction (Second Edition)</i> , Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018	- Implement a simple DQN on a small environment. - Visualize the performance with and without experience replay.
9	Policy Gradient Methods - Introduction to policy gradients and REINFORCE algorithm. - Difference between value-	9	<i>Reinforcement Learning: An Introduction (Second Edition)</i> , Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018	- Implement the REINFORCE algorithm - Experiment with hyperpara

	based and policy-based methods.			meters for stable training.
10	Advanced Topics and Project Showcase - Overview of advanced RL topics: Actor-Critic, PPO, A3C. - Real-world RL applications and ethical considerations.	9	<i>Reinforcement Learning: An Introduction (Second Edition), Authors: Richard S. Sutton and Andrew G. Barto, Publication Year: 2018</i>	- Overview of advanced RL topics: Actor-Critic, PPO, A3C. - Real-world RL applications and ethical considerations.

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignments**: Homework 1 Project 1 Quiz1 Mid-term	50 15 15 20 50	100
2 nd attestation	Assignments**: Homework 2 Project 2 Quiz2 End-term	50 15 15 20 50	100
Final exam*	Test		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

*** The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.*

** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught

25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (50%) and summative rating done during the examination session (50%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.

- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and Slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals

Full name	Job title	Date	Sign
Darkhan Zholtyaev	Assistant Professor	6.12.2024	